Thermodynamics Practice Problems

Problem 1: Reversible Isothermal Expansion of an Ideal Gas

A sealed cylinder with a frictionless piston contains **2.50 mol** of an ideal gas initially at a temperature of **350 K** and a volume of **4.00 L**. The piston is slowly pulled outward while the cylinder remains in contact with a large thermal reservoir, so the gas stays at a constant temperature throughout the expansion. The final volume of the gas is **16.0 L**. Because the temperature is constant, the internal energy of an ideal gas does not change during this process. However, the gas still does work as it pushes the piston outward. Assume the expansion is reversible, meaning the external pressure is always only slightly less than the gas pressure. Use **R = 8.314 J/(mol K)**. Be careful with units and remember that for an isothermal ideal gas process, the work depends on the ratio of the final and initial volumes rather than on the absolute volume values by themselves.

> **Goal: Find the work done by the gas during the isothermal expansion.**

Problem 2: Entropy Change While Heating Water

A container holds **0.750 kg** of liquid water initially at **15.0 degrees C**. The water is slowly heated at constant pressure until it reaches a final temperature of **95.0 degrees C**. Assume the water does not boil during this process, so it remains liquid the entire time. The specific heat capacity of water is approximately constant over this temperature range and is given by **c = 4186 J/(kg K)**. Since the heating is slow, you may treat the process as reversible for the purpose of calculating the entropy change of the water. Remember that entropy calculations involving temperature ratios must use absolute temperature in kelvin, not Celsius. This problem is not asking for the entropy change of the surroundings. It is only asking for the entropy change of the water itself. Pay attention to the fact that the entropy change is not simply Q/T when the temperature changes continuously; instead, you must account for the changing temperature during the heating process.

> **Goal: Calculate the entropy change of the water as it warms from 15.0 degrees C to 95.0 degrees C.**

Problem 3: Reversible Adiabatic Compression of a Diatomic Gas

A piston-cylinder device contains **1.20 mol** of a diatomic ideal gas initially at a temperature of **290 K** and a pressure of **1.00 atm**. The gas is compressed slowly and reversibly until its final volume is **one-fifth** of its initial volume. The cylinder is perfectly insulated, so no heat enters or leaves the gas during the process. Because the compression is reversible and adiabatic, the gas temperature increases according to the adiabatic relation between temperature and volume. For a diatomic ideal gas, use **gamma = 1.40**. You do not need to calculate the initial or final volume explicitly unless you want to check your reasoning. Focus on using the volume ratio correctly, since the final volume is smaller than the initial volume. Since the gas is compressed rather than expanded, the final temperature should be higher than the initial temperature. This problem is mainly testing whether you can connect insulation, reversibility, and the volume ratio to the correct adiabatic temperature relation.

> **Goal: Determine the final temperature of the gas after the adiabatic compression.**